Crazyflie 2.1 Setup Log

Crazyradio 2.0 Ubuntu Setup, Bug Diagnosis and Resolution

Karan Kumar Anand

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1 Objective

The objective of this document is to record the complete setup process of the Crazyradio 2.0 and Crazyflie 2.1 on Ubuntu, including every issue encountered, diagnostic steps performed, root cause analysis, and the final solution.

This document serves as future documentation so that the setup can be reproduced without repeating the debugging process.

2 System Configuration

- Operating System: Ubuntu
- Python Virtual Environment
- cfclient Version: 2026.4
- cflib Version: 0.1.32
- Crazyradio: Crazyradio 2.0
- Drone: Crazyflie 2.1

3 Initial Problem

After installing the Crazyflie client,

- cfclient launched successfully.
- Interface list was empty.
- Scan returned nothing.
- Drone could not be connected.

Symptoms:

- Crazyradio LED was red.
- Crazyflie LEDs were powered.
- No radio interface appeared inside cfclient.

4 Diagnosis Process

4.1 Step 1: Verify USB Detection

Command:

```
lsusb
```

Initially:

```
35f0:bad2 Bitcraze Crazyradio 2.0
```

Observation:
Linux detected the USB device.
Therefore,

- USB port was functional.
- Hardware was detected.

4.2 Step 2: Test cflib

Command

```
python -c "
import cflib.crtp
cflib.crtp.init_drivers()
print(cflib.crtp.scan_interfaces())
"
```

Output

```
Cannot find Crazyradio
[]
```

Conclusion:
The USB device existed but the Crazyradio driver was not being initialized.

4.3 Step 3: Check USB Permissions

Running with sudo removed the permission error.
Therefore Linux permissions were investigated.

4.4 Step 4: Inspect USB Device

Using PyUSB

```
import usb.core
usb.core.find(...)
```

PyUSB successfully detected the device.
Therefore

- USB communication worked.
- Driver issue was elsewhere.

4.5 Step 5: Inspect Mounted Devices

Command

```
lsblk
```

Unexpectedly,

```
/media/.../Crazyradio2
```

appeared.
This indicated that the Crazyradio mounted as a USB storage device.

4.6 Step 6: Inspect Bootloader Files

Contents:

```
CURRENT.UF2
INFO_UF2.TXT
README.TXT
README.HTM
```

INFO_UF2.TXT contained

```
UF2 Bootloader
Bitcraze Crazyradio 2.0
```

Conclusion:
The Crazyradio was booting into UF2 Bootloader instead of application firmware.

5 Root Cause

The Crazyradio application firmware was missing.
The hardware itself was completely functional.
Only the bootloader was running.

6 Solution

Downloaded official firmware
Version:

```
crazyradio2-5.5.uf2
```

Download:

```
wget https://github.com/bitcraze/crazyradio2-firmware/releases/download/5.5/craz
```

Copied firmware

```
cp crazyradio2-5.5.uf2 /media/iiitd/Crazyradio2/
```

After copying,
the Crazyradio rebooted automatically.

7 Verification

Running

```
lsusb
```

now returned

```
1915:7777 Nordic Semiconductor ASA
Bitcraze Crazyradio (PA) dongle
```

This confirmed

- Bootloader exited successfully.
- Application firmware was running.

8 Permission Issue

Next issue:

```
Access denied
(insufficient permissions)
```

Solution:
Created

```
/etc/udev/rules.d/99-bitcraze.rules
```

Contents

```
SUBSYSTEM=="usb",
ATTRS{idVendor}=="1915",
ATTRS{idProduct}=="7777",
MODE="0666"

SUBSYSTEM=="usb",
ATTRS{idVendor}=="35f0",
ATTRS{idProduct}=="bad2",
MODE="0666"

SUBSYSTEM=="usb",
ATTRS{idVendor}=="0483",
ATTRS{idProduct}=="5740",
MODE="0666"
```

Reloaded rules

```
sudo udevadm control --reload-rules
sudo udevadm trigger
```

After reconnecting the Crazyradio,
permission issues disappeared.

9 Driver Verification

Running

```
python - <<EOF
import cflib.crtp
cflib.crtp.init_drivers()
print(cflib.crtp.get_interfaces_status())
EOF
```

Output

```
{
'radio':'Crazyradio version 5.4',
'UsbCdc':'No information available',
'udp':'No information available',
'prrt':'No information available',
'cpx':None
}
```

Meaning

- Crazyradio firmware operational.
- USB communication working.
- cflib successfully communicating with radio.

10 Final Connection

Initially,
Scan still returned

```
[]
```

Reason:
The Crazyflie was not powered on.
After powering the Crazyflie,
cfclient immediately detected

```
radio://0/80/2M
```

Connection became successful.
Battery voltage

```
4.017 V
```

Link quality was displayed and telemetry started updating.

11 Lessons Learned

1. If the Crazyradio mounts as a USB drive named Crazyradio2, it is in UF2 bootloader mode.
2. The bootloader itself is not an error.
3. Flashing the official firmware restores normal operation.
4. Linux USB detection alone does not imply application firmware is running.
5. Always verify using

   ```
   lsusb
   ```

   Expected

   ```
   1915:7777
   ```

6. Verify driver initialization using

   ```
   cflib.crtp.get_interfaces_status()
   ```

   Expected

   ```
   Crazyradio version 5.4
   ```

7. If Scan returns an empty list, first verify that the Crazyflie itself is powered.

12 Current Status

- Crazyradio firmware restored.
- Ubuntu configured successfully.
- USB permissions fixed.
- cfclient installed.
- cflib installed.
- Crazyradio detected.
- Crazyflie connected successfully.
- Telemetry operational.
- Ready for development and experimentation.